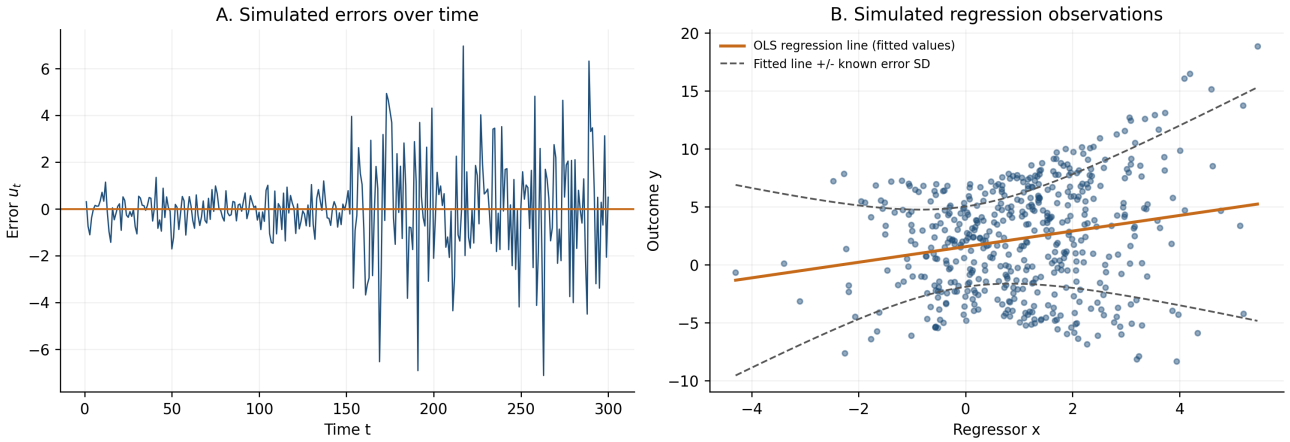
III Tutorial

Heteroskedasticity: graphical evidence and robust inference

Exercise 1. Recognising heteroskedasticity

Use the regression model $y_i = \beta_0 + \beta_1 x_i + u_i$, where u_i is the **error term**. After estimating OLS, the **fitted values** are $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ and the **residuals** are $\widehat{u}_i = y_i - \widehat{y}_i$. Residuals estimate the unobserved errors.

Panel A plots simulated errors over time, with an orange zero reference line. Panel B uses the simulated sample from Monte_Carlo_Simulation.ipynb, with an orange **OLS regression line** showing fitted values. The vertical distance from each observation to that line is its residual. Dashed curves add and subtract one known error standard deviation to illustrate dispersion; they are not confidence intervals.



To do.

(a) Describe how the vertical spread around the orange reference line changes in each panel. What does this suggest about error dispersion?

(b) Which OLS assumption does each pattern call into question? Does either pattern necessarily imply a failure of the zero conditional mean assumption for errors? Explain why a sloping regression line does not itself imply nonzero expected errors.

(c) Does the pattern in A by itself establish serial correlation? Explain why graphical evidence should be interpreted alongside the model assumptions.

Relevant concepts

- Errors; fitted values; residuals; zero conditional mean assumption for errors.
- Homoskedasticity and heteroskedasticity; error variance versus fitted values.
- Changing dispersion over time or with a regressor; heteroskedasticity versus serial correlation.

Exercise 2. Heteroskedasticity: income and food expenditure

Notebook: Heteroskedasticity.ipynb

Analyse annual income and food expenditure for 235 Belgian households in 1857, both measured in Belgian francs. Estimate a simple OLS regression, inspect residual dispersion, and compare conventional and HC3 inference on the same sample. The zoomed plot changes only the display; all estimates use all 235 households.

Relevant concepts

- Conditional mean versus conditional variance; homoskedasticity and heteroskedasticity.
- Residual plots; interpretation of dispersion at comparable income levels.
- Conventional and robust standard errors; t-tests; p-values; confidence intervals; statistical versus economic significance.
- Zero conditional mean; unbiasedness versus validity of conventional inference.

To do. Using the estimated income slope, calculate the fitted change in annual food expenditure for a 250-franc increase in income. Convert the conventional and HC3 95% slope intervals into intervals for this fitted change.

Exercise 3. OLS assumptions: theoretical DGP

Consider a simple regression with an intercept:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \qquad u_i = U_i\sqrt{4 + x_i^2}, \qquad i = 1, \ldots, n.$$

The observations are i.i.d., with $n \geq 3$, $x_i \sim N(1, 2)$ (mean 1, variance 2), and $U_i \sim \text{Uniform}(-3, 3)$ independent of x_i.

To do.

(a) Derive $E[u_i \mid x_i]$. Does zero conditional mean hold? Explain the implication for the unbiasedness of OLS.

(b) Derive $\text{Var}(u_i \mid x_i)$. Is the error homoskedastic? Explain the consequences for conventional OLS standard errors and inference.

Exercise 4. Monte Carlo simulation: OLS inference (optional)

Notebook: Monte_Carlo_Simulation.ipynb

Use the DGP in Exercise 3 with $\beta_0 = 1$ and $\beta_1 = 0.5$. Compare samples of **250 and 1,000 observations**, with **10,000 replications per size**. Examine the OLS sampling distribution and compare conventional, HC1 and HC3 standard errors, interval widths and 95% coverage. All methods use the same samples and an approximate t_{n-2} reference.

Relevant concepts

- Sampling distribution; unbiasedness; empirical sampling SD versus estimated SE.
- Confidence-interval coverage; significance level; Type I and Type II errors; empirical rejection rate under a true null.
- Robust inference in finite and large samples; sample size versus replications; Monte Carlo uncertainty.

Solutions

Exercise 1. Recognising heteroskedasticity

(a) Error and residual dispersion

In A, errors fluctuate around the zero reference line, with much greater dispersion in the second half. In B, the orange OLS regression line gives fitted values $\widehat{y}_i$; the vertical distances $y_i - \widehat{y}_i$ are residuals $\widehat{u}_i$. Their spread is smallest near $x = 0$ and grows with $|x|$, suggesting changing error variance. This vertical spread is different from the horizontal spread of the regressor.

(b) OLS assumptions about the errors

Both panels illustrate a failure of constant error variance, or **heteroskedasticity**. This is distinct from the **zero conditional mean assumption for errors**, $E[u_i \mid x_i] = 0$: errors can average to zero while their dispersion changes. Both simulations are constructed with zero-mean errors. In B, increasing fitted values reflect the positive estimated slope; they do not imply nonzero expected errors. The dashed curves illustrate error dispersion using the known simulation standard deviations, not uncertainty about the fitted regression line.

If the error term has zero conditional mean, heteroskedasticity alone does not make OLS coefficients biased, under independent sampling and the usual rank and integrability conditions. Conventional standard errors, p-values and confidence intervals can nevertheless be misleading.

(c) Heteroskedasticity versus serial correlation

The pattern in A does not by itself establish serial correlation. The simulation uses independent, zero-mean errors with a change in scale: heteroskedasticity does not require serial correlation. Serial correlation concerns covariance between errors at different dates, rather than a change in their variance. For empirical data, visual patterns suggest hypotheses; they do not establish the model assumptions. In particular, OLS residuals sum to zero when an intercept is included; this sample property does not prove $E[u_i \mid x_i] = 0$ for the unobserved errors.

Exercise 3. OLS assumptions: theoretical DGP

(a) Zero conditional mean and unbiasedness

For $U_i \sim \text{Uniform}(-3, 3)$,

$$E[U_i] = \frac{-3+3}{2} = 0.$$

Conditioning on x_i fixes $\sqrt{4+x_i^2}$. Independence gives $E[U_i \mid x_i] = E[U_i]$, so

$$\begin{aligned} E[u_i \mid x_i] &= E\left[U_i\sqrt{4+x_i^2} \mid x_i\right] \\ &= \sqrt{4+x_i^2}\, E[U_i \mid x_i] \\ &= \sqrt{4+x_i^2} \times 0 = 0. \end{aligned}$$

Thus zero conditional mean holds. Let $X = (x_1, \ldots, x_n)$. The i.i.d. sampling assumption also gives $E[u_i \mid X] = 0$. Using the OLS slope expression,

$$\widehat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_i (x_i - \bar{x})u_i}{\frac{1}{n}\sum_i (x_i - \bar{x})^2}.$$

The denominator is positive with probability one for this continuous regressor. Conditioning on X then gives

$$E[\widehat{\beta}_1 \mid X] = \beta_1 + \frac{\frac{1}{n}\sum_i (x_i - \bar{x})E[u_i \mid X]}{\frac{1}{n}\sum_i (x_i - \bar{x})^2} = \beta_1.$$

Iterated expectations imply $E[\widehat{\beta}_1] = \beta_1$. The intercept is also unbiased, since $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x}$ and $E[\bar{u} \mid X] = 0$. Homoskedasticity is not needed for unbiasedness.

(b) Conditional variance and inference

The variance of a uniform variable is

$$\text{Var}(U_i) = \frac{(3-(-3))^2}{12} = 3.$$

Using independence and the variance rule for a fixed multiplier,

$$\begin{aligned} \text{Var}(u_i \mid x_i) &= \text{Var}\left(U_i\sqrt{4+x_i^2} \mid x_i\right) \\ &= (4+x_i^2)\,\text{Var}(U_i \mid x_i) \\ &= \boxed{3(4+x_i^2)}. \end{aligned}$$

This variance changes with x_i^2: it is 12 at $x_i = 0$ and 24 at $x_i = 2$. The errors are therefore **heteroskedastic**, with dispersion increasing with $|x_i|$, even though their conditional mean is zero.

Conventional standard errors, p-values and confidence intervals can be misleading. Robust standard errors estimate uncertainty allowing for heteroskedasticity; they leave the OLS coefficients and fitted values unchanged. Their inference is approximate in finite samples, and their validity here relies on independent observations and zero conditional mean.